

CS 422 – CS303 Language Theory & Finite Automata

Project Assignment 2025-2026

Real-Life Application of Automata Theory

Project Title: /Smart Home Security System Using Face Recognition/

Group Members:	Student Name	Student ID
	1- Jana Nawaf shammary	2240006642
	2- Marwa Abdulaziz	2240000811
	3- Alya shehab	2240002062
	4- Maha Aljoaib	2240006849
	5- Maisa Alzahrani	2240004660
Submission Date:	29/03/2026	
Instructor:	[Lina Awad]	

1.1 Real-Life Application

1.1.1 Application Choice

Selected Application: Smart Home Security System Using Face Recognition

This project models a smart home security system that uses facial recognition technology to control access to a house. The system identifies authorized users and allows them to enter only after their identity has been verified.

1.1.2 System Description

A smart home security system is designed to enhance safety by controlling access to a house using facial recognition technology. When a person approaches the door, a camera detects the presence of a face and scans it. The system then verifies the identity using a facial recognition system that compares the scanned face with stored authorized users in the system database. If the identity is confirmed, the door unlocks automatically and the user is allowed to enter the home. Otherwise, access is denied and the door remains locked. This process follows a specific sequence of operations, which makes it suitable to model using concepts from Automata Theory.

1.1.3 Alphabet Definition

$\Sigma = \{\text{detect_face, scan_face, verify_face, unlock_door, enter_home}\}$

detect_face → the system detects a person near the door

scan_face → the system scans the detected face

verify_face → the system verifies whether the user is authorized

unlock_door → the system unlocks the door

enter_home → the user enters the home

1.1.4 Language Description

$L = \{\text{detect_face scan_face verify_face unlock_door enter_home}\}$

The language represents all valid sequences of operations that allow a user to successfully access the smart home. A valid sequence must follow the correct order of detecting a face, scanning it, verifying the identity, unlocking the door, and finally allowing the user to enter the home.

1.1.5 Valid Example (Member of Language)

Example string that BELONGS to the language:

detect_face scan_face verify_face unlock_door enter_home

This sequence follows the correct order of operations in the smart home system. The system first detects a face, then scans it, verifies the identity, unlocks the door, and finally allows the authorized user to enter the home.

1.1.6 Invalid Example (Not a Member of Language)

Example string that DOES NOT belong to the language:

scan_face verify_face unlock_door enter_home

This sequence is invalid because the system attempts to scan a face before detecting it. The correct process must start with the detect_face operation.

1.2 Simplified Version

1.2.1 Simplified System Description

To simplify the system for modeling purposes, we reduce the number of operations to only the essential steps required for accessing the home. In the simplified model, the system detects a face, verifies the identity, unlocks the door, and allows the user to enter the home.

1.2.2 Simplified Alphabet

$\Sigma_{\text{simplified}} = \{d, v, u, e\}$

Where:

d = detect face

v = verify face

u = unlock door

e = enter home

1.2.3 Simplified Language Description

$L_{\text{simplified}} = \{d v u e\}$

The simplified language represents valid sequences of operations in the simplified smart home security system. A correct sequence begins with detecting a face, followed by verifying the identity, unlocking the door, and finally allowing the user to enter the home.

1.2.4 Valid Example (Simplified Version)

d v u e

This sequence follows the correct order of operations in the simplified system.

1.2.5 Invalid Example (Simplified Version)

v d u e

This sequence is invalid because the system cannot verify a face before detecting it.